

HAT-P-54b

R-1812

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_161209 · created 2026-08-03T03:39:30+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457731.84903 +/- 0.00043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.199 +/- 0.011
Transit depth (Rp/Rs)^2	3.97 +/- 0.45 [%]
Orbital Inclination (inc)	86.01 +/- 1.35
Airmass coefficient 1 (a1)	1.194 +/- 0.02
Airmass coefficient 2 (a2)	-0.164 +/- 0.041
Scatter in the residuals of the lightcurve fit is	1.64 %
Best Comparison Star	None
Optimal Aperture	3.39
Optimal Annulus	11.31
Transit Duration (day)	0.0665 +/- 0.0068

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.199 ± 0.011 vs 0.1572 (deviation 3.8σ)
Duration fit vs published	0.0665 d vs 0.075 d (deviation 1.25σ)
Mid-time O–C	-0.8 min
Depth SNR	18.1
Residual scatter	1.64 %

Light curve

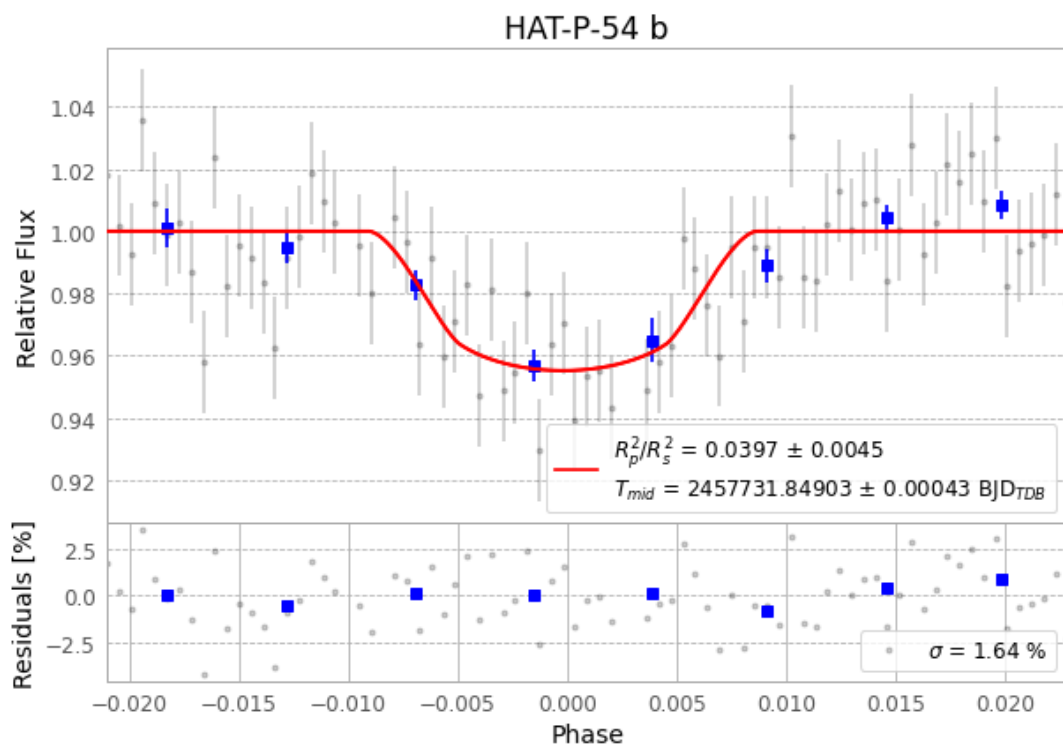


Figure 1 — FinalLightCurve_HAT-P-54 b_2016-12-08.png (SHA-256 a49f42f511db1f30...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [152, 257], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f4c80ccade0446a4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

81 FITS frames (53 MB), HATP-54161209061812.FITS → HATP-54161209101812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-161209.log (e83053bf7c81...), FinalLightCurve_HAT-P-54 b_2016-12-08.png (a49f42f511db...), FinalLightCurve_HAT-P-54 b_2016-12-08.csv (434ede47e2ca...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 03:39Z.